

ICT Trading Bot: Forex Edition

OANDA / MT5 CSV (also Binance USD-M) · branch `claude/keen-meitner-hlq76o` · 4 Oct 2026

1. Bottom line

Backtested on one year of real EUR/USD (15 Oct 2025 to 30 Sep 2026, 0.8-pip spread), the strategy as specced has **no edge**. The bias is right about as often as a coin flip, and the entry only makes money when the direction is already right. **Do not trade it live**. Three fixes tried (bias lock time, 4H structure, a 20-day trend filter) did not change that. 11 unit tests pass; the OANDA connection and order placement are untested.

2. Pipeline

Step	What it does
1. Daily bias	Locked once per day at 02:00 New York (London open), from data before the lock only. Votes (+1 / -1): A price vs previous-day high/low, B 4H swing structure (HH+HL / LH+LL), C yesterday's liquidity sweep. Trade only when $ \text{score} \geq 2$. A fourth vote D (price vs 20-day average of closes) is reported alone and as an optional filter, not part of the combined score. Buys on bullish days, sells on bearish days, nothing on neutral days.
2. 15m sweep	After the lock: wick beyond the lowest low (bullish) or highest high (bearish) of the previous 24 15m candles, with a close back inside.
3. 5m IFVG	Track opposite-type FVGs. A candle <i>body</i> closing through the gap inverts it (wicks do not count), after the sweep candle closes.
4. Entry	Retest of the inverted zone with a rejection candle in the bias direction, inside a kill zone (02:00-05:00 or 07:00-10:00 NY).
5. Risk	Stop 2 pips beyond the sweep extreme; stops under 5 pips skipped. Target 1:3 R:R. One trade per day; open trades close at the day's end.

3. Forex handling

Area	Behaviour
Trading day	17:00 to 17:00 New York, DST-aware. Weekend stubs (fewer than 100 5m candles) are skipped.
Data	Only 5m candles needed (1m exports are resampled). 15m, 4H and daily are built from them on the same day boundaries.
Costs	--spread-pips is charged per trade in R. Pip size inferred from the symbol (JPY 0.01, gold 0.1, else 0.0001).
Commands	<code>backtest --csv f.csv, backtest --source oanda, signal, live</code> (OANDA practice by default; <code>--execute</code> to send orders). Options: <code>--killzones</code> , <code>--lock-offset</code> , <code>--loose-body</code> , <code>--warmup</code> .

4. Backtest: EUR/USD, 15 Oct 2025 to 30 Sep 2026

Default settings (kill zones on, strict gap-body rule). 'Direction right' = price moved the bias way from the lock to the day's close.

Bias method	Bias days	Direction right	Trades	Win rate	Total R
A: previous-day high/low	49	51.0%	8	25.0%	-1.85
B: 4H structure	153	52.9%	46	17.4%	-18.78
B: daily structure (old)	137	46.0%	40	22.5%	-19.82
C: liquidity sweep	64	50.0%	15	13.3%	-11.04
Combined (2+ votes)	17	47.1%	6	0.0%	-4.55

With kill zones off and the loose body rule, every method still lost after spread (A -0.9R, B 4H -12.6R, C -13.4R, combined -4.7R). Before the fix the old setup had method B's direction right only 41.6% of the time.

5. Diagnostic: is it the bias or the entry?

Entry logic with no bias filter (loose body, kill zones off). Hindsight direction cannot be traded; it only shows what the entry needs.

Direction used	Trades	Win rate	Total R
Hindsight (actual day direction)	129	55.8%	+42.8
Wrong way	209	14.4%	-158.5
Either direction, first setup	233	27.9%	-86.1

The entry is roughly break-even with no filter, so all the value must come from the bias, and a bias right about 53% of the time does not deliver it. Break-even at 1:3 is a 25% win rate before costs; with stops of 5 to 9 pips, a 0.8-pip spread costs about 0.1 to 0.15R per trade.

6. 20-day moving-average trend filter

Tested 3 Nov 2025 to 30 Sep 2026 (20 days of warm-up so the average always has data). '+sma20' keeps a method's bias only when it agrees with the trend vote. Kill zones on, strict body unless noted.

Bias	Direction right	Trades	Win rate	Total R
D: trend vote alone	48.9%	59	23.7%	-18.1
B: 4H structure	53.5%	43	18.6%	-16.6
B: 4H structure + sma20	52.0%	31	16.1%	-14.1
Combined	53.3%	5	0.0%	-3.5
Combined + sma20	61.5%	4	0.0%	-2.4
Kill zones off, loose body:				
D: trend vote alone	48.9%	141	37.6%	-2.6
B: 4H structure	53.5%	93	31.2%	-10.5
B: 4H structure + sma20	52.0%	65	32.3%	-5.1
Combined + sma20	61.5%	7	14.3%	-1.0

The trend vote called direction right only 48.9% of the time. Gating trims losses only because it takes fewer trades; no row is clearly positive. Split at 1 Apr 2026 (kill zones off, loose body), the trend vote alone made -19.8R in the first half and +17.2R in the second, and combined + sma20 went from +1.9R (2 trades) to -2.9R (5 trades). That instability is what noise looks like, and the best-looking rows rest on 2 to 4 trades.

7. Caveats and next steps

- One pair, one year, in which EUR/USD fell from 1.17 to 1.135. Test other years and pairs before drawing wider conclusions.
- Method C is implemented as 'closed back beyond the swept level' rather than the spec's `c Lose > d[-2] . low`. 'Body fully through' means the body spans the whole gap.
- MT5 exports are usually in broker time, not UTC; convert before use or day boundaries and kill zones will be off.

- Next steps: test other pairs (GBP/USD, USD/JPY) and earlier years; try a stop of 10 pips or more or a next-liquidity-pool target; stop tuning filters on this one year, since more tweaks would mostly fit noise.

Appendix: source (ict_fvg_bot.py)

```
#!/usr/bin/env python3
"""ICT bot for forex (OANDA / MT5 CSV) and Binance USD-M futures. Standard library only.

Pipeline: daily bias -> 15m liquidity sweep -> 5m IFVG retest -> entry (1:3 R:R), inside a kill zone.

Forex trading day = 17:00 New York to 17:00 New York (DST-aware); bias is locked at that open.
Crypto trading day = 00:00 UTC. Only 5m data is needed: 15m and daily candles are built from it
with the same day boundaries, so the data source's own daily alignment never matters.

# backtest on MT5/broker CSV export (no network needed)
python ict_fvg_bot.py backtest --market forex --symbol EUR_USD --csv EURUSD_M5.csv --start 2025-01-01 --end 2026-01-01 --
# backtest straight from OANDA (needs OANDA_TOKEN)
python ict_fvg_bot.py backtest --source oanda --symbol EUR_USD --start 2025-01-01 --end 2026-01-01
# today's bias / setups
python ict_fvg_bot.py signal --source oanda --symbol EUR_USD
# live loop on an OANDA practice account (dry-run unless --execute)
python ict_fvg_bot.py live --source oanda --symbol EUR_USD --risk-amount 50
"""
from __future__ import annotations

import argparse
import csv
import hashlib
import hmac
import json
import os
import time
import urllib.parse
import urllib.request
from bisect import bisect_left
from dataclasses import dataclass, field
from datetime import datetime, timedelta, timezone
from zoneinfo import ZoneInfo

MIN = 60_000
HOURL = 3_600_000
BULL, BEAR = 1, -1
UTC, NY = ZoneInfo("UTC"), ZoneInfo("America/New_York")

# ----- markets
@dataclass(frozen=True)
class Market:
    name: str
    tz: ZoneInfo
    anchor_hour: int # local hour at which the trading day starts
    killzones: tuple = () # ((start_hr, end_hr), ...) New York time; () = always allowed
    min_candles: int = 100 # 5m candles for a day to count (skips weekends/holidays)
    lock_offset: int = 0 # hours after the day opens at which the bias is locked

    def day_start(self, t):
        local = datetime.fromtimestamp(t / 1000, self.tz)
        d = (local - timedelta(hours=self.anchor_hour)).date()
        return int(datetime(d.year, d.month, d.day, self.anchor_hour, tzinfo=self.tz).timestamp() * 1000)

    def day_end(self, start):
        d = datetime.fromtimestamp(start / 1000, self.tz).date() + timedelta(days=1)
        return int(datetime(d.year, d.month, d.day, self.anchor_hour, tzinfo=self.tz).timestamp() * 1000)

    def lock_time(self, start):
        """Bias lock instant: `lock_offset` wall-clock hours after the day opens (DST-safe)."""
        return int((datetime.fromtimestamp(start / 1000, self.tz) + timedelta(hours=self.lock_offset)).timestamp() * 1000)

    def in_killzone(self, t):
        if not self.killzones:
            return True
        h = datetime.fromtimestamp(t / 1000, NY)
        x = h.hour + h.minute / 60
        return any(a <= x < b for a, b in self.killzones)

FOREX = Market("forex", NY, 17, killzones=((2, 5), (7, 10)), lock_offset=9) # lock 02:00 NY (London open)
CRYPTO = Market("crypto", UTC, 0)
MARKETS = {"forex": FOREX, "crypto": CRYPTO}

def pip_size(symbol):
    s = symbol.upper()
    return 0.01 if "JPY" in s else 0.1 if s.startswith("XAU") else 0.0001
@dataclass
```

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class Candle:
    t: int # open time, ms
    o: float
    h: float
    l: float
    c: float

# ----- resampling
def resample(cs, key_fn, end_fn=None, until=None):
    """Group consecutive candles by key_fn(t). Drops a final bucket that ends after `until` (still forming)."""
    out, last = [], None
    for c in cs:
        k = key_fn(c.t)
        if k != last:
            out.append(Candle(k, c.o, c.h, c.l, c.c)); last = k
        else:
            b = out[-1]; b.h = max(b.h, c.h); b.l = min(b.l, c.l); b.c = c.c
    if until is not None and out and end_fn(out[-1].t) > until:
        out.pop()
    return out

def to_15m(c5, until=None):
    return resample(c5, lambda t: t - t % (15 * MIN), lambda k: k + 15 * MIN, until)

def to_h4(c5, market, until=None):
    """4H candles aligned to the market's day open (forex: 17,21,01,05,09,13 NY)."""
    def key(t):
        ds = market.day_start(t)
        return ds + (t - ds) // (4 * HOUR) * 4 * HOUR
    return resample(c5, key, lambda k: k + 4 * HOUR, until)

def to_daily(c5, market, until=None):
    """Daily candles on the market's day boundaries; thin days (weekend/holiday stubs) are dropped."""
    counts = {}
    for c in c5:
        k = market.day_start(c.t); counts[k] = counts.get(k, 0) + 1
    days = resample(c5, market.day_start, market.day_end, until)
    return [d for d in days if counts[d.t] >= market.min_candles]

# ----- sources
def _http(url, params=None, headers=None, method="GET", body=None):
    if params:
        url += "?" + urllib.parse.urlencode(params)
    req = urllib.request.Request(url, headers=headers or {}, method=method,
                                data=json.dumps(body).encode() if body is not None else None)
    with urllib.request.urlopen(req, timeout=20) as r:
        return json.load(r)

def load_csv(path):
    """5m candles from a CSV export (MT5 / broker / generic). Columns by header name, else positional.

    Accepted time forms: epoch seconds/ms, ISO-8601, 'YYYY.MM.DD HH:MM[:SS]' (MT5), or separate Date and Time
    columns. Times are UTC. Header is optional; positional order is time,open,high,low,close.
    """
    with open(path, newline="") as f:
        sample = f.read(4096); f.seek(0)
        dialect = csv.Sniffer().sniff(sample, delimiters=";\t")
        rows = [r for r in csv.reader(f, dialect) if r]
        head = [x.strip().lower().strip("<>") for x in rows[0]]
        named = "open" in head
        if named:
            col = {n: head.index(n) for n in ("open", "high", "low", "close")}
            tcols = [head.index(n) for n in ("date", "time") if n in head] or [head.index("timestamp") if "timestamp" in head else 0]
            rows = rows[1:]
        else:
            col, tcols = {"open": 1, "high": 2, "low": 3, "close": 4}, [0]
        out = []
        for r in rows:
            ts = ".join(r[i].strip() for i in tcols)
            out.append(Candle(_parse_time(ts), float(r[col["open"]]), float(r[col["high"]]), float(r[col["low"]]), float(r[col["c"]]))
            out.sort(key=lambda c: c.t)
        if len(out) > 10 and sorted(b.t - a.t for a, b in zip(out, out[1:]))[len(out) // 2] == MIN: # 1m export -> 5m
            out = resample(out, lambda t: t - t % (5 * MIN))
        return out

def _parse_time(s):
    s = s.strip()

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if s.endswith("+00:00"):
    s = s[:-6]
if s.replace(".", "", 1).isdigit() and "." not in s[:5]:
    v = float(s)
    return int(v if v > 1e11 else v * 1000)
for fmt in ("%Y.%m.%d %H:%M:%S", "%Y.%m.%d %H:%M", "%Y-%m-%d %H:%M:%S", "%Y-%m-%d %H:%M", "%Y-%m-%dT%H:%M:%S", "%Y-%m-%dT"):
    try:
        return int(datetime.strptime(s[:19] if "T" in s else s, fmt).replace(tzinfo=UTC).timestamp() * 1000)
    except ValueError:
        pass
raise ValueError(f"unrecognised time: {s!r}")

class Binance:
    name = "binance"

    def __init__(self, testnet=False):
        self.base = "https://testnet.binancefuture.com" if testnet else "https://fapi.binance.com"
        self.key, self.secret = os.environ.get("BINANCE_API_KEY", ""), os.environ.get("BINANCE_API_SECRET", "")

    def fetch_5m(self, symbol, start, end):
        out, cur = [], start
        while cur < end:
            rows = _http(self.base + "/fapi/v1/klines", {"symbol": symbol, "interval": "5m", "startTime": cur, "endTime": end})
            out += [Candle(r[0], float(r[1]), float(r[2]), float(r[3]), float(r[4])) for r in rows]
            if len(rows) < 1500:
                break
            cur = rows[-1][0] + 1
        return out

    def place(self, symbol, su, a):
        q = urllib.parse.urlencode
        def signed(path, params):
            params = dict(params, timestamp=int(time.time() * 1000), recvWindow=5000)
            qs = q(params)
            sig = hmac.new(self.secret.encode(), qs.encode(), hashlib.sha256).hexdigest()
            return _http(f"{self.base}{path}?{qs}&signature={sig}", headers={"X-MBX-APIKEY": self.key}, method="POST")
        info = _http(self.base + "/fapi/v1/exchangeInfo")
        step = next(float({x["filterType"]: x for x in s["filters"]}[["LOT_SIZE"]][["stepSize"]]) for s in info["symbols"] if s["symbol"] == symbol)
        qty = int(a.risk_amount / abs(su.entry - su.stop) / step) * step
        side, opp = ("BUY", "SELL") if su.direction == BULL else ("SELL", "BUY")
        common = dict(symbol=symbol, quantity=f"{qty:.8f}".rstrip("0"))
        signed("/fapi/v1/order", dict(common, side=side, type="MARKET"))
        signed("/fapi/v1/order", dict(common, side=opp, type="STOP_MARKET", stopPrice=f"{su.stop:.2f}", reduceOnly="true"))
        signed("/fapi/v1/order", dict(common, side=opp, type="TAKE_PROFIT_MARKET", stopPrice=f"{su.target:.2f}", reduceOnly="true"))

class Oanda:
    """OANDA v20. Env: OANDA_TOKEN, OANDA_ACCOUNT. Practice account unless env='live'."""
    name = "oanda"

    def __init__(self, env="practice"):
        self.base = "https://api-fxpractice.oanda.com" if env == "practice" else "https://api-fxtrade.oanda.com"
        self.token, self.account = os.environ.get("OANDA_TOKEN", ""), os.environ.get("OANDA_ACCOUNT", "")
        self.h = {"Authorization": f"Bearer {self.token}", "Content-Type": "application/json"}

    def fetch_5m(self, symbol, start, end):
        out, cur = [], start
        while cur < end:
            d = _http(f"{self.base}/v3/instruments/{symbol}/candles", {"granularity": "M5", "price": "M", "from": f"{cur / 1000:.0f}", "count": 5000, self.h})
            rows = [c for c in d["candles"] if c["complete"]]
            if not d["candles"]:
                break
            for c in rows:
                t = int(datetime.strptime(c["time"][:19], "%Y-%m-%dT%H:%M:%S").replace(tzinfo=UTC).timestamp() * 1000)
                if t < end:
                    m = c["mid"]; out.append(Candle(t, float(m["o"]), float(m["h"]), float(m["l"]), float(m["c"])))
            last = int(datetime.strptime(d["candles"][-1]["time"][:19], "%Y-%m-%dT%H:%M:%S").replace(tzinfo=UTC).timestamp())
            if len(d["candles"]) < 5000 or last + 5 * MIN <= cur:
                break
            cur = last + 5 * MIN
        return out

    def place(self, symbol, su, a):
        pip = a.pip or pip_size(symbol)
        dec = len(f"{pip:.10f}".rstrip("0").split(".")[1]) + 1
        units = int(a.risk_amount / (abs(su.entry - su.stop) * a.quote_rate)) * su.direction
        order = {"type": "MARKET", "instrument": symbol, "units": str(units), "timeInForce": "FOK", "positionFill": "DEFAULT", "stopLossOnFill": {"price": f"{su.stop:.{dec}f"}}, "takeProfitOnFill": {"price": f"{su.target:.{dec}f}"}}
        return _http(f"{self.base}/v3/accounts/{self.account}/orders", headers=self.h, method="POST", body={"order": order})

# ----- swings / FVG logic
def swing_highs(cs, n=2):

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    return [i for i in range(n, len(cs) - n)
            if all(cs[i].h > cs[i - k].h and cs[i].h >= cs[i + k].h for k in range(1, n + 1))]

def swing_lows(cs, n=2):
    return [i for i in range(n, len(cs) - n)
            if all(cs[i].l < cs[i - k].l and cs[i].l <= cs[i + k].l for k in range(1, n + 1))]

@dataclass
class Zone:
    kind: int
    low: float
    high: float
    formed: int
    inverted_at: int = -1

def fvg_at(cs, i):
    """FVG completed by candle i (pattern i-2, i-1, i), or None."""
    if i < 2:
        return None
    if cs[i].l > cs[i - 2].h:
        return Zone(BULL, cs[i - 2].h, cs[i].l, i)
    if cs[i].h < cs[i - 2].l:
        return Zone(BEAR, cs[i].h, cs[i - 2].l, i)
    return None

def inverts(zone, c, strict=True):
    """Candle body closes through the gap (wicks don't count). strict: body spans the whole gap."""
    if zone.kind == BEAR:
        return c.c > zone.high and (not strict or c.o <= zone.low)
    return c.c < zone.low and (not strict or c.o >= zone.high)

# ----- daily bias
def vote_pdh_pdl(d, price):
    return BULL if price > d[-1].h else BEAR if price < d[-1].l else 0

def vote_structure(d, n=2):
    hi, lo = swing_highs(d, n)[-2:], swing_lows(d, n)[-2:]
    if len(hi) < 2 or len(lo) < 2:
        return 0
    up = d[hi[1]].h > d[hi[0]].h and d[lo[1]].l > d[lo[0]].l
    dn = d[hi[1]].h < d[hi[0]].h and d[lo[1]].l < d[lo[0]].l
    return BULL if up else BEAR if dn else 0

def vote_sweep(d, lookback=5):
    if len(d) < lookback + 1:
        return 0
    prev, y = d[-lookback - 1:-1], d[-1]
    lo, hi = min(c.l for c in prev), max(c.h for c in prev)
    if y.l < lo and y.c > lo:
        return BULL
    if y.h > hi and y.c < hi:
        return BEAR
    return 0

# Votes take (daily history, 4H history, price at lock). COMBINED are the methods summed for the bias;
# B_structure_1d is reported for comparison only.
def vote_trend(d, price, n=20):
    """D. Price above the n-day simple average of daily closes +1, below -1 (needs n completed days)."""
    if len(d) < n:
        return 0
    ma = sum(c.c for c in d[-n:]) / n
    return BULL if price > ma else BEAR if price < ma else 0

METHODS = {"A_pdh_pdl": lambda d, h, p: vote_pdh_pdl(d, p),
           "B_structure_4h": lambda d, h, p: vote_structure(h),
           "B_structure_1d": lambda d, h, p: vote_structure(d),
           "C_sweep": lambda d, h, p: vote_sweep(d),
           "D_sma20": lambda d, h, p: vote_trend(d, p)}
COMBINED = ("A_pdh_pdl", "B_structure_4h", "C_sweep")

def daily_bias_votes(d, price, h4=None):
    return {k: f(d, h4 if h4 is not None else d, price) for k, f in METHODS.items()}

def bias_from_votes(votes, min_score=2):
    score = sum(votes[k] for k in COMBINED)

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return BULL if score >= min_score else BEAR if score <= -min_score else 0

def lock_bias(daily, h4_all, c5, ds, p):
    """Bias for the day opening at ds, using only data before the lock instant. None if no price at lock."""
    lock_t = p.market.lock_time(ds)
    t5 = [c.t for c in c5]
    i = bisect_left(t5, lock_t) - 1 # last 5m candle that closed by the lock
    if i < 0 or c5[i].t < lock_t - 30 * MIN or c5[i].t < ds:
        if p.market.lock_offset or bisect_left(t5, lock_t) >= len(c5) or c5[bisect_left(t5, lock_t)].t > lock_t + 30 * MIN:
            return None
        i = bisect_left(t5, lock_t) # lock at the open: price = first candle's open
        votes = daily_bias_votes(daily, c5[i].o, [c for c in h4_all if c.t + 4 * HOUR <= lock_t])
        return bias_from_votes(votes), votes, lock_t
    h4 = h4_all[:bisect_left([c.t for c in h4_all], lock_t - 4 * HOUR + 1)] # H4 candles completed by lock
    votes = daily_bias_votes(daily, c5[i].c, h4)
    return bias_from_votes(votes), votes, lock_t

# ----- 15m sweep -> 5m IFVG entry
@dataclass
class Setup:
    direction: int
    entry_t: int
    entry: float
    stop: float
    target: float
    sweep_extreme: float
    sweep_t: int
    zone: Zone = field(repr=False, default=None)

@dataclass
class Params:
    market: Market = FOREX
    sweep_lookback: int = 24
    fvg_wait: int = 48
    rr: float = 3.0
    pip: float = 0.0001
    buffer_pips: float = 2.0 # forex: stop distance beyond the sweep extreme
    buffer_frac: float = 0.0005 # crypto: same, as a fraction of price
    min_risk_pips: float = 5.0 # forex: skip trades whose stop is tighter than this (spread eats them)
    spread_pips: float = 0.0 # backtest cost per trade
    strict_body: bool = True
    max_trades_per_day: int = 1

    def buffer(self, price):
        return self.buffer_pips * self.pip if self.market.name == "forex" else price * self.buffer_frac

    def min_risk(self):
        return self.min_risk_pips * self.pip if self.market.name == "forex" else 0.0

def find_sweeps(c15, k0, k1, direction, p):
    for k in range(max(k0, p.sweep_lookback), k1):
        win, c = c15[k - p.sweep_lookback:k], c15[k]
        if direction == BULL:
            lvl = min(x.l for x in win)
            if c.l < lvl and c.c > lvl:
                yield k, lvl, c.l
        else:
            lvl = max(x.h for x in win)
            if c.h > lvl and c.c < lvl:
                yield k, lvl, c.h

def find_ifvg_entry(c5, i0, i1, direction, sweep_t, p):
    """FVG -> body-close inversion (after the sweep closes) -> retest rejection inside the kill zone."""
    sweep_close = sweep_t + 15 * MIN
    wanted = BEAR if direction == BULL else BULL
    active, inverted = [], []
    for j in range(i0, min(i1, i0 + p.fvg_wait)):
        c = c5[j]
        for z in inverted[:]:
            if direction == BULL:
                if c.c < z.low:
                    inverted.remove(z); continue
                if c.l <= z.high and c.c > z.high and c.c > c.o and p.market.in_killzone(c.t):
                    return c, z
            else:
                if c.c > z.high:
                    inverted.remove(z); continue

```

```

        if c.h >= z.low and c.c < z.low and c.c < c.o and p.market.in_killzone(c.t):
            return c, z
    if c.t >= sweep_close:
        for z in active[:]:
            if inverts(z, c, p.strict_body):
                active.remove(z); z.inverted_at = j; inverted.append(z)
    z = fvg_at(c5, j)
    if z and z.kind == wanted and c5[j - 2].t >= sweep_t:
        active.append(z)
return None

def find_setups(c15, c5, day_start, day_end, direction, p, after=0):
    """Setups for one trading day, earliest first. direction = locked daily bias."""
    if direction == 0:
        return []
    t15, t5 = [c.t for c in c15], [c.t for c in c5]
    k0, k1 = bisect_left(t15, max(day_start, after)), bisect_left(t15, day_end)
    i_end = bisect_left(t5, day_end)
    out, last_entry_t = [], -1
    for k, lvl, extreme in find_sweeps(c15, k0, k1, direction, p):
        sweep_t = c15[k].t
        hit = find_ifvg_entry(c5, bisect_left(t5, sweep_t), i_end, direction, sweep_t, p)
        if not hit:
            continue
        c, z = hit
        if c.t <= last_entry_t:
            continue
        entry, buf = c.c, p.buffer(c.c)
        stop = extreme - buf if direction == BULL else extreme + buf
        risk = (entry - stop) * direction
        if risk <= 0 or risk < p.min_risk():
            continue
        out.append(Setup(direction, c.t, entry, stop, entry + direction * p.rr * risk, extreme, sweep_t, z))
        last_entry_t = c.t
    out.sort(key=lambda s: s.entry_t)
    return out

def simulate(setup, c5, day_end):
    """(R multiple before costs, reason). Same-candle SL+TP -> SL. Open trades close at the day's end."""
    d, risk, last = setup.direction, (setup.entry - setup.stop) * setup.direction, None
    t5 = [c.t for c in c5]
    for c in c5[bisect_left(t5, setup.entry_t + 5 * MIN):]:
        if c.t >= day_end:
            break
        last = c
        if (c.l <= setup.stop) if d == BULL else (c.h >= setup.stop):
            return -1.0, "SL"
        if (c.h >= setup.target) if d == BULL else (c.l <= setup.target):
            return (setup.target - setup.entry) * d / risk, "TP"
    if last is None:
        return 0.0, "NOFILL"
    return (last.c - setup.entry) * d / risk, "EOD"

# ----- backtest
def backtest(daily, h4, c15, c5, p, start_ms=0, end_ms=None, warmup=12, fee_r=0.0):
    """Same pipeline gated by each bias method alone and by the combined score.

    Bias is locked at market.lock_time() from data before that instant only; setups must come after it.
    'dir acc' = did price move in the bias direction from the lock price to the day's close.
    """
    variants = list(METHODS) + ["combined"]
    gated = [v + "+sma20" for v in variants if v != "D_sma20"] # bias kept only if it agrees with the trend
    stats = {v: dict(bias_days=0, bias_right=0, trades=0, wins=0, r=0.0) for v in variants + gated}
    log, t5 = [], [c.t for c in c5]
    for i in range(warmup, len(daily)):
        today = daily[i]
        if today.t < start_ms or (end_ms and today.t >= end_ms):
            continue
        end = p.market.day_end(today.t)
        lk = lock_bias(daily[:i], h4, c5, today.t, p)
        if lk is None:
            continue
        comb, votes, lock_t = lk
        j = bisect_left(t5, lock_t) - 1
        actual = BULL if today.c > c5[j].c else BEAR
        biases = dict(votes, combined=comb)
        for v in variants:
            if v != "D_sma20":
                biases[v + "+sma20"] = biases[v] if biases[v] == votes["D_sma20"] else 0

```

```

    for v in variants + gated:
        b, s = biases[v], stats[v]
        if b == 0:
            continue
        s["bias_days"] += 1; s["bias_right"] += (b == actual)
        for su in find_setups(c15, c5, today.t, end, b, p, after=lock_t)[: p.max_trades_per_day]:
            r, why = simulate(su, c5, end)
            r -= fee_r + p.spread_pips * p.pip / ((su.entry - su.stop) * su.direction)
            s["trades"] += 1; s["wins"] += r > 0; s["r"] += r
            if v == "combined":
                log.append((datetime.fromtimestamp(su.entry_t / 1000, UTC), b, su.entry, su.stop, su.target, why, r))
    return stats, log

def print_report(stats, log):
    print(f"{'bias method':<16}{{'bias days':>10}{'dir acc':>9}{'trades':>8}{'win%':>7}{'total R':>9}{'avg R':>8}}")
    for v, s in stats.items():
        acc = s["bias_right"] / s["bias_days"] * 100 if s["bias_days"] else 0
        wr = s["wins"] / s["trades"] * 100 if s["trades"] else 0
        avg = s["r"] / s["trades"] if s["trades"] else 0
        print(f"{'v':<16}{s['bias_days']:>10}{acc:>8.1f}%{s['trades']:>8}{wr:>6.1f}%{s['r']:>9.2f}{avg:>8.2f}")
    print("\ncombined-score trades (UTC):")
    for t, b, e, sl, tp, why, r in log:
        print(f" {t:%Y-%m-%d %H:%M} {'LONG ' if b == BULL else 'SHORT'} entry={e:.5f} sl={sl:.5f} tp={tp:.5f} {why:<5} {r:+.

# ----- data + live
def get_broker(a):
    return Oanda(getattr(a, "env", "practice")) if a.source == "oanda" else Binance(getattr(a, "testnet", False))

def make_params(a):
    m = MARKETS[a.market]
    if getattr(a, 'lock_offset', None) is not None:
        m = Market(m.name, m.tz, m.anchor_hour, m.killzones, m.min_candles, a.lock_offset)
    if a.killzones is not None:
        kz = tuple(tuple(float(x) for x in z.split("-")) for z in a.killzones.split(",")) if a.killzones else ()
        m = Market(m.name, m.tz, m.anchor_hour, kz, m.min_candles, m.lock_offset)
    p = Params(market=m, rr=a.rr, pip=a.pip or pip_size(a.symbol), spread_pips=a.spread_pips,
               strict_body=not a.loose_body, min_risk_pips=a.min_risk_pips)
    return p

def day_state(broker, symbol, p, now_ms):
    """Today's context from completed candles only: (daily history, 4H, 15m, 5m, day_start)."""
    ds = p.market.day_start(now_ms)
    raw = broker.fetch_5m(symbol, ds - 40 * 24 * HOUR, now_ms)
    c5 = [c for c in raw if c.t + 5 * MIN <= now_ms]
    until = c5[-1].t + 5 * MIN if c5 else None
    daily = to_daily([c for c in c5 if c.t < ds], p.market)
    return daily, to_h4(c5, p.market, until), to_15m(c5, until), c5, ds

BIAS_NAME = {BULL: "BULLISH", BEAR: "BEARISH", 0: "NEUTRAL"}

def run_live(a):
    p, broker = make_params(a), get_broker(a)
    locked, traded = {}, set()
    while True:
        now = int(time.time() * 1000)
        daily, h4, c15, c5, ds = day_state(broker, a.symbol, p, now)
        lock_t = p.market.lock_time(ds)
        if now >= lock_t and ds not in locked: # lock exactly once per day
            lk = lock_bias(daily, h4, c5, ds, p)
            if lk:
                locked[ds] = lk[0]
                print(f"[{datetime.fromtimestamp(lock_t / 1000, UTC):%Y-%m-%d %H:%M}Z] bias locked {BIAS_NAME[lk[0]]} {lk[1]}")
        if ds in locked:
            for su in find_setups(c15, c5, ds, p.market.day_end(ds), locked[ds], p, after=lock_t)[: p.max_trades_per_day]:
                if su.entry_t in traded or su.entry_t < now - 10 * MIN:
                    continue
                traded.add(su.entry_t)
                print("SIGNAL", su)
                if a.execute:
                    print(broker.place(a.symbol, su, a))
            time.sleep(30)

def ts(s):
    return int(datetime.strptime(s, "%Y-%m-%d").replace(tzinfo=UTC).timestamp() * 1000)
def main():

```

```

ap = argparse.ArgumentParser(description=__doc__, formatter_class=argparse.RawDescriptionHelpFormatter)
sub = ap.add_subparsers(dest="cmd", required=True)
def common(sp):
    sp.add_argument("--market", choices=list(MARKETS), default=None, help="default: forex for oanda/csv, crypto for binan")
    sp.add_argument("--source", choices=["oanda", "binance"], default="oanda")
    sp.add_argument("--symbol", default="EUR_USD", help="OANDA EUR_USD style; Binance BTCUSDT")
    sp.add_argument("--pip", type=float, default=0.0, help="override pip size")
    sp.add_argument("--rr", type=float, default=3.0)
    sp.add_argument("--spread-pips", type=float, default=0.0)
    sp.add_argument("--min-risk-pips", type=float, default=5.0)
    sp.add_argument("--killzones", default=None, help="NY-time windows e.g. '2-5,7-10'; '' disables")
    sp.add_argument("--lock-offset", type=int, default=None, help="hours after the day opens to lock bias (forex default")
    sp.add_argument("--loose-body", action="store_true", help="IFVG needs only a close beyond the gap")
    sp.add_argument("--env", choices=["practice", "live"], default="practice", help="OANDA environment")
    sp.add_argument("--testnet", action="store_true", help="Binance testnet")
b = sub.add_parser("backtest"); common(b)
b.add_argument("--csv", help="5m CSV (MT5/broker export, UTC); skips the network")
b.add_argument("--start", required=True); b.add_argument("--end", required=True)
b.add_argument("--fee-r", type=float, default=0.0)
b.add_argument("--warmup", type=int, default=12, help="days of history before the first traded day")
s = sub.add_parser("signal"); common(s)
l = sub.add_parser("live"); common(l)
l.add_argument("--execute", action="store_true", help="place orders (default: log signals only)")
l.add_argument("--risk-amount", type=float, default=10.0, help="account-currency risk per trade")
l.add_argument("--quote-rate", type=float, default=1.0, help="account currency per 1 unit of quote currency")
a = ap.parse_args()
a.market = a.market or ("crypto" if a.source == "binance" else "forex")
p = make_params(a)

if a.cmd == "backtest":
    s0, e0 = ts(a.start), ts(a.end)
    if a.csv:
        c5 = load_csv(a.csv)
    else:
        c5 = get_broker(a).fetch_5m(a.symbol, s0 - 45 * 24 * HOUR, e0)
    c5 = [c for c in c5 if c.t < e0 + 2 * 24 * HOUR]
    stats, log = backtest(to_daily(c5, p.market), to_h4(c5, p.market), to_15m(c5), c5, p, start_ms=s0, end_ms=e0, warmup=
    print(f"{a.symbol} {a.start}..{a.end} market={a.market} pip={p.pip} spread={p.spread_pips}p "
          f"killzones={p.market.killzones or 'off'} strict_body={p.strict_body} candles5m={len(c5)}")
    print_report(stats, log)
elif a.cmd == "signal":
    now = int(time.time() * 1000)
    daily, h4, c15, c5, ds = day_state(get_broker(a), a.symbol, p, now)
    lk = lock_bias(daily, h4, c5, ds, p) if now >= p.market.lock_time(ds) else None
    if not lk:
        return print("bias not locked yet (before the lock time or no price at lock)")
    print("votes:", lk[1], "->", BIAS_NAME[lk[0]])
    for su in find_setups(c15, c5, ds, p.market.day_end(ds), lk[0], p, after=lk[2]):
        print(su)
else:
    run_live(a)

if __name__ == "__main__":
    main()

```